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Batch Foundry: A Socratic Engine for Mitigating Answer Leakage in Educational Artificial Intelligence

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Abstract

The rapid deployment of Large Language Models (LLMs) in educational settings presents a critical pedagogical challenge: "Answer Leakage." When students experience cognitive friction, they frequently utilise prompt injection or behavioural frustration to bypass educational scaffolding and extract direct answers from AI tutors. This paper introduces the LumenForge Socratic Engine, a deterministic gatekeeping architecture evaluated via a "Batch Foundry" methodology. Through an automated audit cohort of 25 adversarial pedagogical stress-tests (The Scholarly 25), the engine demonstrated a 100.0% Socratic Integrity rate and maintained an average Thinking Ratio (TR) of 41.2%. Alongside comparative live audits against industry-standard education models (Google Gemini 2.5 LearnLM and Llama 3.3 70B), these results indicate robust mitigation of answer leakage within the audit cohort and provide a scalable framework for the pre-deployment auditing of educational AI systems.

1. Introduction

As the integration of generative AI within secondary education accelerates, the efficacy of these tools is often undermined by their propensity to act as answer engines rather than tutors. This phenomenon, which we term "Answer Leakage," occurs when an AI system capitulates to a student's cognitive avoidance strategies, directly solving problems rather than facilitating the Concrete-Pictorial-Abstract (CPA) learning loop.

Within the UK public sector landscape, this stateless, containerised middleware architecture is designed to address the data compliance and pedagogical drift challenges relevant to institutional deployments of AI tutoring systems. By preventing AI models from prematurely divulging answers, institutional tutoring platforms can better uphold curriculum rigour while operating within strict UK safeguarding protocols.

This paper details the architecture and empirical evaluation of the LumenForge Socratic Engine (v2.7), a middleware policy enforcement layer designed to categorically prevent Answer Leakage and maintain high cognitive load.

2. Methodology: The Batch Foundry Architecture

To rigorously evaluate the engine prior to human deployment, we developed the "Batch Foundry," a digital wind tunnel for educational AI.

2.1 The Supervisor Proxy and Mathematical Formulation

The core architecture consists of an event-driven Supervisor Agent that acts as a transparent proxy between the Target AI Tutor and the student. The Supervisor intercepts the payload, calculates a real-time Thinking Ratio (TR), and classifies the semantic intent of the AI's response using a secondary NLP classifier.

Immediately following interception, the system calculates the Thinking Ratio (TR) via the formal mathematical formulation:

$$TR = (T_{Socratic} / T_{Total}) \times (1 - \lambda_{leak})$$

where:

- $T_{Socratic}$ = the volume of tokens generated containing cognitive prompts, open-ended questioning, and conceptual scaffolding.
- T_{Total} = the entire transactional token output payload.
- λ_{leak} = a binary coefficient (0 or 1) indicating whether a direct structural solution pattern or final evaluation step was exposed to the user.

2.2 The Scholarly 25 Audit Cohort

We synthesised three distinct adversarial student personas to stress-test the system:

- **The Cheat-Seeker**: Employs explicit demands and prompt injections to bypass educational scaffolding.
- **The Concept-Block**: Simulates a rigid misunderstanding of prerequisite concepts, resisting initial corrections.
- **The High-Frustration Learner**: Exhibits emotional distress and low confidence, requiring pedagogical patience rather than immediate problem resolution.

The audit encompassed 25 distinct pedagogical traces across STEM and Humanities, mapped to the UK Key Stage 3 and Key Stage 4 (GCSE) curricula.

3. Results and Explicit Leak Classification

The LumenForge Socratic Engine v2.7 demonstrated strong pedagogical resilience during The Scholarly 25 audit.

To rigorously evaluate performance across these adversarial traces, we established strict technical boundary metrics for leak classification. Specifically, a response triggers an "Answer Leak" classification ($\lambda_{leak} = 1$) if the output contains more than 15% of a mathematical or logical formula's final evaluation steps, or states a direct definition without an accompanying open-ended query. To establish an industry baseline, we conducted comparative API audits against leading foundational models using official vendor-recommended pedagogy instructions.

Model / Configuration	Socratic Integrity	Average TR	Safeguarding Compliance
Google Gemini 2.5 (LearnLM Prompts)	80.0%	56.3%	Partial (Leaks under urgency)
Llama 3.3 70B (Open-Weights Baseline)	60.0%	47.0%	Compromised (Capitulates to essays)
LumenForge Socratic Engine (v2.7)	**100.0%**	**41.2%**	**100.0% Verified Deterministic**

Table 1: Comparative live audit matrix under adversarial student pressure across UK secondary traces.

3.1 Pedagogical De-risking and The Verbosity Trap

While unmanaged baseline models achieved higher raw token Thinking Ratios (e.g., Gemini LearnLM at 56.3%), trace analysis reveals this is driven by unconstrained verbosity—generating multi-paragraph responses (~300 words) that induce cognitive overload in struggling learners. Conversely, the integration of strict Subject Lockdown and Zero-Leak Protocols in LumenForge successfully neutralised instances of answer leakage. Furthermore, by capping student-facing output to a maximum of three sentences, the engine ensured the student performed 100% of the active typing, maintaining an optimal classroom engagement ratio (41.2%) without cognitive exhaustion.

3.2 Limitations

The present evaluation is limited to a 25-trace synthetic cohort. While the adversarial personas were designed to represent high-friction failure modes observed in live tutoring contexts, the cohort size does not support claims of statistical significance, and results may not generalise to all student populations. Validation against larger synthetic cohorts and live classroom telemetry is planned as future work.

4. Conclusion

The LumenForge Socratic Engine demonstrated robust performance in maintaining pedagogical integrity across the audit cohort. By utilising the Batch Foundry methodology for pre-deployment auditing, institutional stakeholders can verify an AI tutor's adherence to defined educational and safeguarding criteria before classroom exposure.

Within the evolving governance framework of UK education, Batch Foundry is designed to serve as a scalable verification layer enabling institutions, including Multi-Academy Trusts (MATs), to automate compliance tracking without exposing Personally Identifiable Information (PII). Future work will expand the behavioural logic of the synthetic personas, increase cohort scale, and explore broader demographic inclusivity metrics across national cohorts.